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(54) **SYSTEM OF MEDICAL DEVICES AND METHOD FOR PERICARDIAL PUNCTURE**

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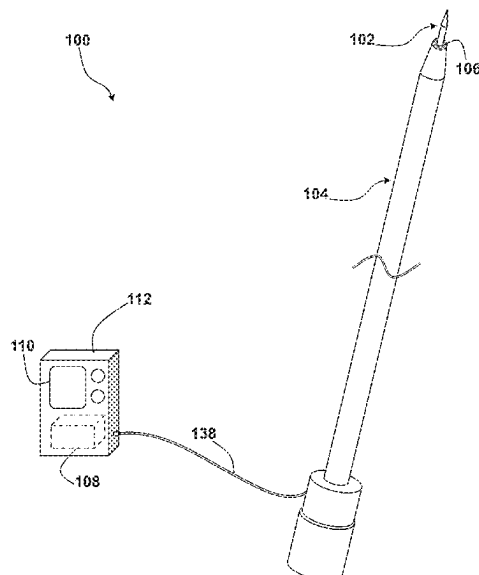
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(57)

ABSTRACT

A system for pericardial puncture includes a medical device having an elongate shaft extending between a proximal end and a distal end, and having a distally facing electrode at the distal end of the shaft. An impedance meter is electrically connectable with the electrode for measuring an impedance associated with the electrode. A display is provided for displaying an indication of a position of the electrode. The indication of the position is based on of the impedance.

14 Claims, 4 Drawing Sheets



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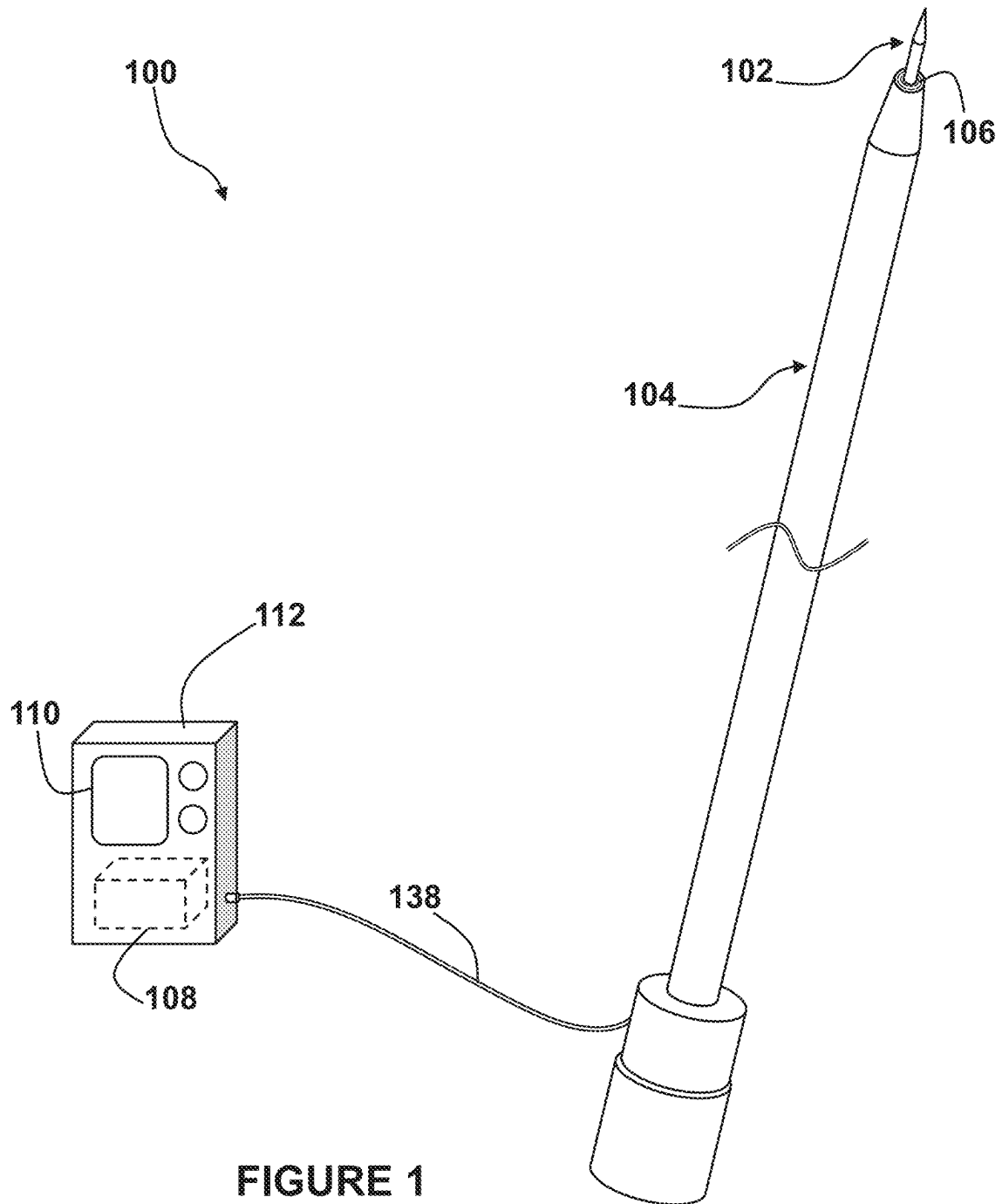
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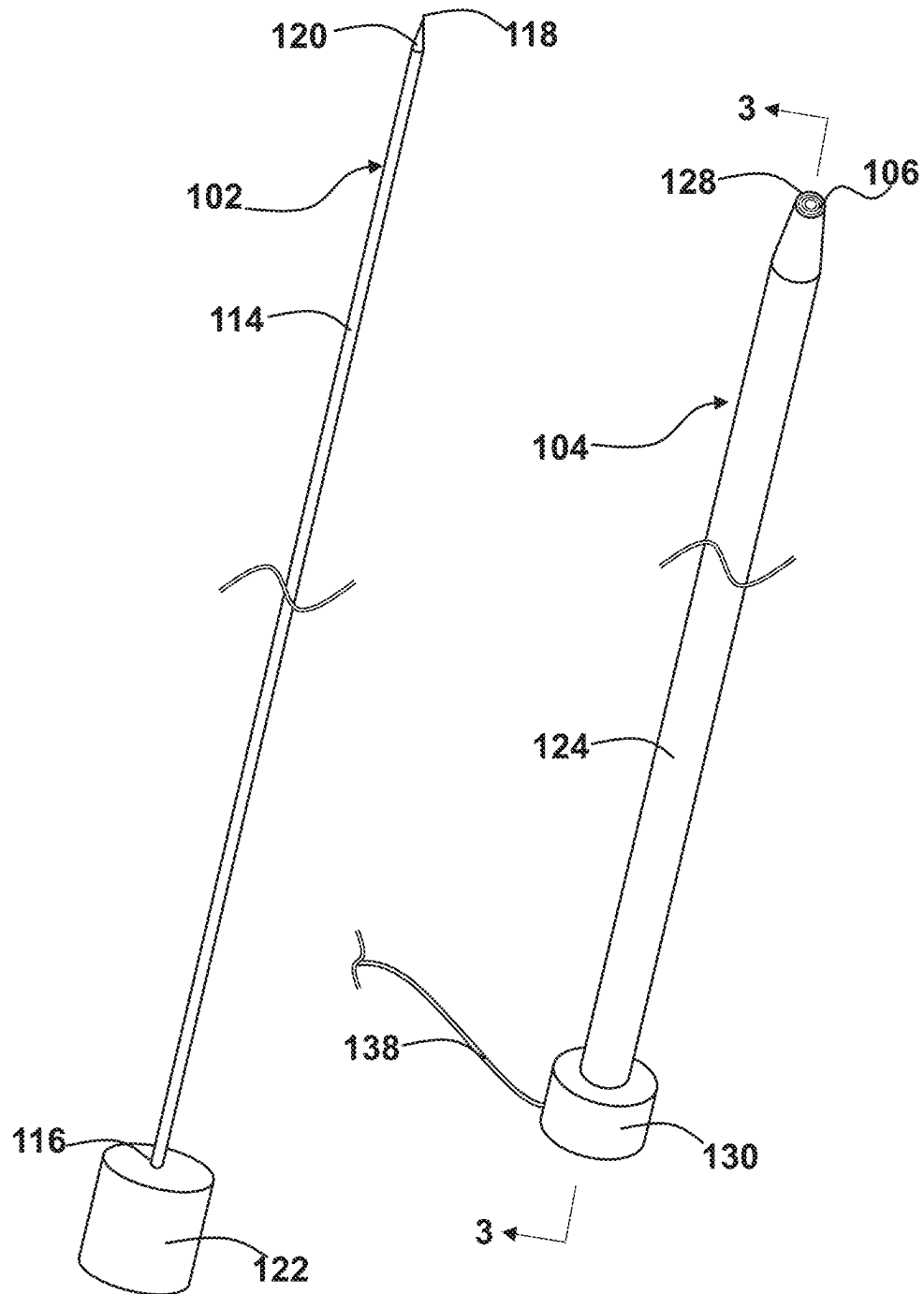


FIGURE 2

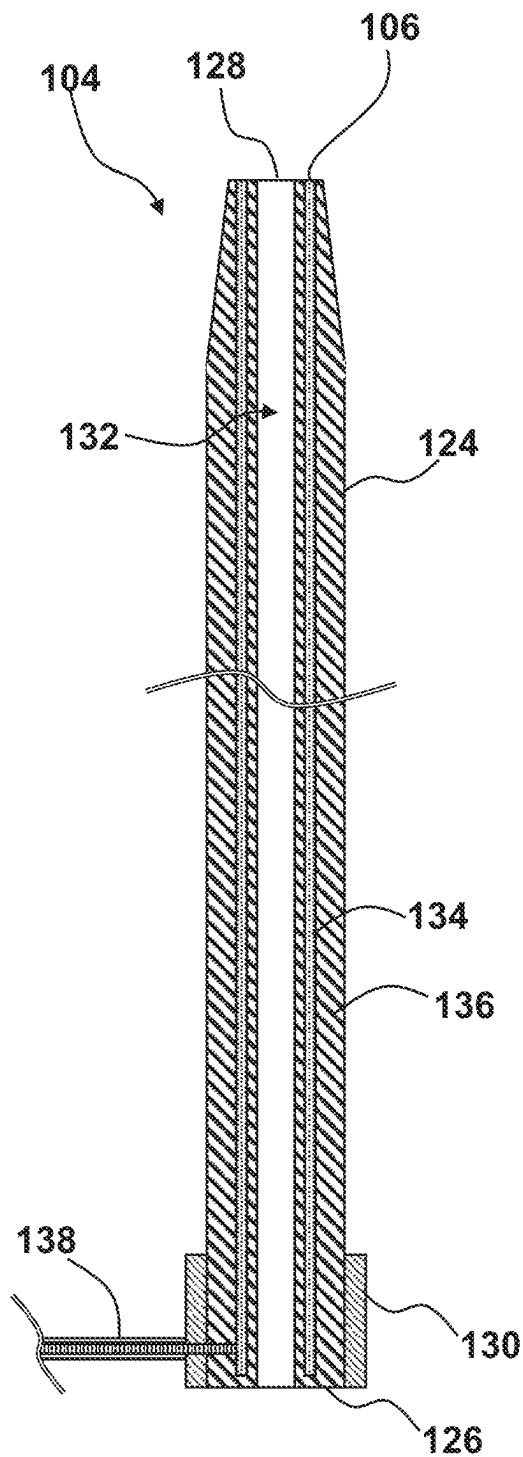


FIGURE 3

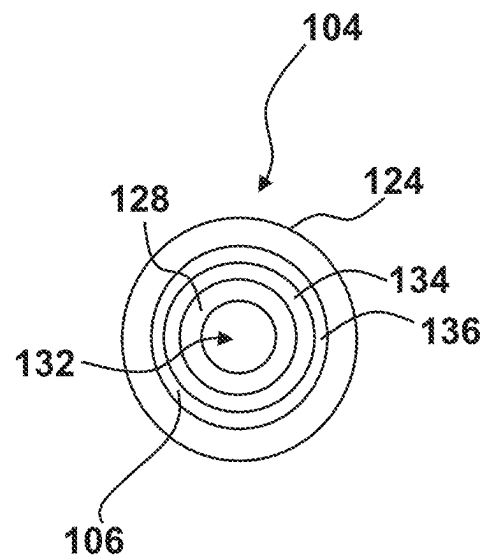
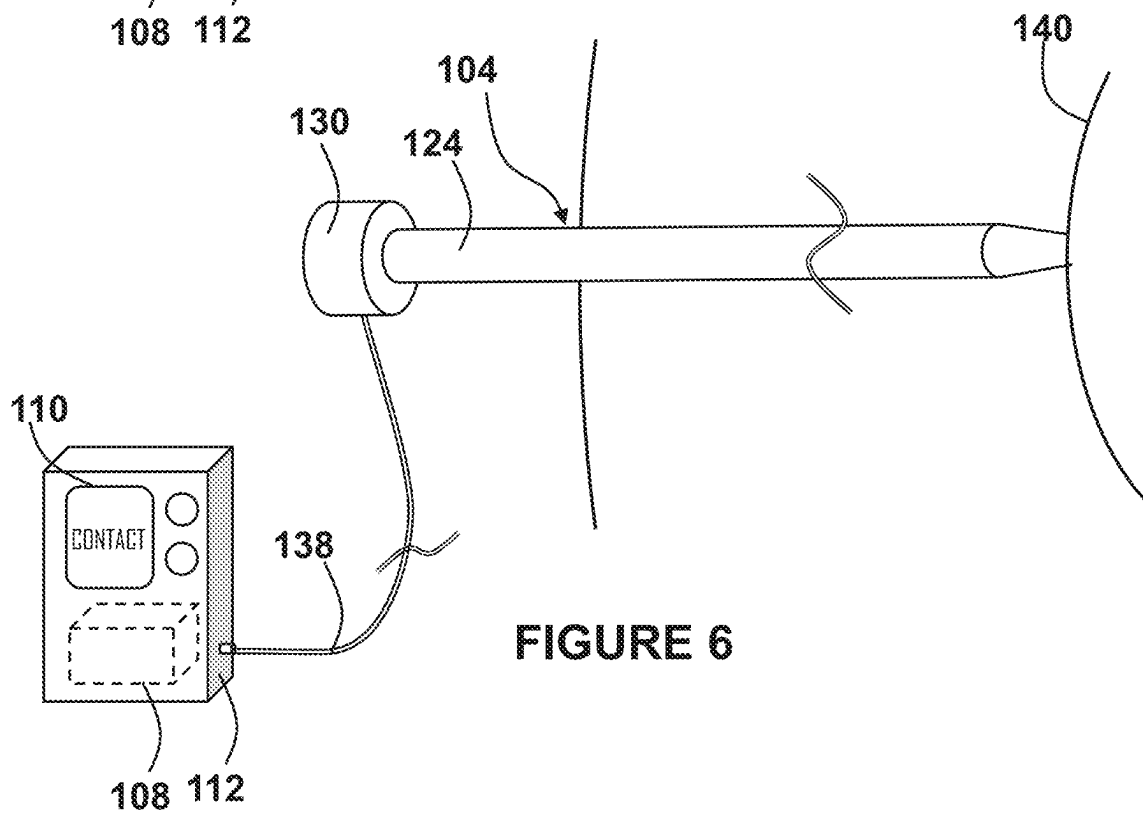
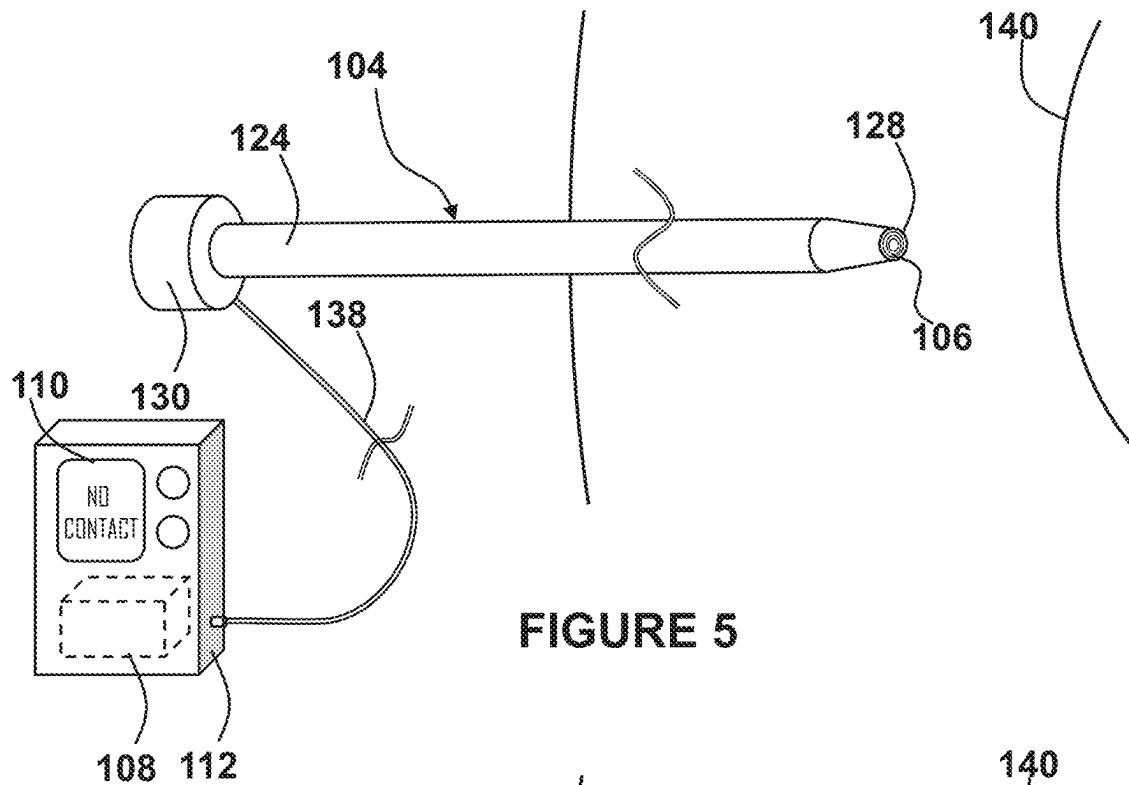


FIGURE 4



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SYSTEM OF MEDICAL DEVICES AND METHOD FOR PERICARDIAL PUNCTURE

FIELD

This document relates to medical devices. More specifically, this document relates to systems of medical devices that can be used in pericardial puncture, and related methods.

SUMMARY

The following summary is intended to introduce the reader to various aspects of the detailed description, but not to define or delimit any invention.

Systems for pericardial puncture are disclosed. According to some aspects, a system for pericardial puncture includes a medical device having an elongate shaft extending between a proximal end and a distal end, and having an electrode at the distal end of the shaft. An impedance meter is electrically connectable with the electrode for measuring an impedance associated with the electrode. A display is provided for displaying an indication of a position of the medical device. The indication of the position is based on of the impedance.

In some examples, the indication of the position of the medical device is an indication of whether the medical device is in contact with a target tissue.

In some examples, the medical device is an introducer and the shaft has a lumen extending therethrough from the proximal end to the distal end. The shaft can include a metallic tube and an electrically insulative sheathing on the tube. The electrode can include an electrically exposed section of the tube. The shaft can include an electrical conductor extending proximally from the electrode for electrically connecting the electrode to the impedance meter.

In some examples, the system further includes a puncture device advanceable through the lumen from the proximal end to the distal end. The puncture device can be a radiofrequency puncture device and the system can further include a radiofrequency generator electrically connectable with the electrode for delivering radiofrequency energy to the electrode.

In some examples, the medical device is a puncture device. The system can further include a radiofrequency generator electrically connectable with the electrode for delivering radiofrequency energy from the electrode to puncture tissue. Alternatively, the puncture device can include a sharp tip.

Methods for pericardial puncture are also disclosed. According to some aspects, a method for pericardial puncture includes: a. advancing an introducer towards a pericardium; b. advancing a puncture device through the introducer towards the pericardium; c. puncturing the pericardium with the puncture device; and d. during at least one of step a., step b., and step c., measuring an impedance to assess a position of the introducer and/or a position of the puncture device.

In some examples, step d. is carried out during step a., and the impedance is an impedance associated with an electrode at a distal end of the introducer. In step d., the position of the distal end of the introducer can be assessed. Step d. can include determining whether the distal end of the introducer is in contact with the pericardium.

In some examples, step d. is carried out during step c., and the impedance is an impedance associated with a puncturing tip of the puncture device. In step d., the position of the

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puncturing tip of the puncture device can be assessed. Step d. can include determining whether the puncturing tip has punctured the pericardium.

In some examples, the method further includes: e. displaying an indication of the position of the introducer and/or the position of the puncture device. Step e. can include displaying an indication of whether the introducer has contacted the pericardium. Step e. can include displaying an indication of whether the puncture device has punctured the pericardium.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings are for illustrating examples of articles, methods, and apparatuses of the present disclosure and are not intended to be limiting. In the drawings:

FIG. 1 is a perspective view of an example system of medical devices in an assembled state;

FIG. 2 is a perspective view of the puncture device and introducer of the system of FIG. 1, in an unassembled state.

FIG. 3 is a cross-section taken along line 3-3 in FIG. 2;

FIG. 4 is an end view of the introducer of FIG. 2;

FIG. 5 is a schematic drawing showing a step of a method for pericardial puncture; and

FIG. 6 is a schematic drawing showing a step subsequent to that of FIG. 5.

DETAILED DESCRIPTION

Various apparatuses or processes or compositions will be described below to provide an example of an embodiment of the claimed subject matter. No example described below limits any claim and any claim may cover processes or apparatuses or compositions that differ from those described below. The claims are not limited to apparatuses or processes or compositions having all of the features of any one apparatus or process or composition described below or to features common to multiple or all of the apparatuses or processes or compositions described below. It is possible that an apparatus or process or composition described below is not an embodiment of any exclusive right granted by issuance of this patent application. Any subject matter described below and for which an exclusive right is not granted by issuance of this patent application may be the subject matter of another protective instrument, for example, a continuing patent application, and the applicants, inventors or owners do not intend to abandon, disclaim or dedicate to the public any such subject matter by its disclosure in this document.

Generally disclosed herein is a system that can be used in pericardial puncture procedures. The system is configured to provide an indication of a position of a medical device of the system, and more particularly, to provide an indication that a medical device of the system is in contact with a target tissue (e.g. the pericardium of the heart) or has punctured the target tissue. For example, the system can include an introducer that is percutaneously advanced towards the heart, and the system can provide an indication of whether the distal end of the introducer is in contact with the heart (and thus advancement can be stopped). In order to provide an indication of the position of the medical device, the medical device can include an electrode that is positioned to contact or approach the heart. The system can further include an impedance meter for measuring an impedance associated with the electrode (i.e. for measuring the impedance of an electrical circuit of which the electrode is a part). As the medical device is advanced towards the heart, the impedance

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can be monitored. When the electrode contacts or approaches the heart, the impedance will change, and this change can provide an indication that the medical device is in contact with the heart. The system can further include a display connected to the impedance meter for displaying an indication of the position of the introducer. For example, based on the impedance measured by the impedance meter, the display can be a visual display that displays text or symbols to indicate that the medical device is in contact with the heart, or an auditory display that generates a sound to indicate that the medical device is in contact with the heart. In an alternative example, the system can include a puncture device that is advanced towards the heart through an introducer and then is used to puncture the pericardium, and the system can provide an indication of whether the distal end of the puncture device is in contact with the heart or whether the distal end of the puncture device has passed through the pericardium (and thus advancement can be stopped).

Referring now to FIG. 1, an example system 100 of medical devices is shown. The system 100 generally includes a pair of medical devices, namely a puncture device 102 and an introducer 104. The puncture device 102 is receivable in and advanceable through the introducer 104. The introducer 104 can serve to atraumatically guide the puncture device 102 towards a target location in a patient's body (e.g. the heart), and the puncture device 102 can then puncture a target tissue at the target location (e.g. puncture the pericardium). As will be described in further detail below, the introducer 104 includes an electrode 106, and the system further includes an impedance meter 108 (shown in dotted line in FIG. 1) that is electrically connectable with the electrode 106 for measuring an impedance associated with the electrode 106. The impedance meter 108 can be any device that measures impedance, for example an ohm-meter or an LCR meter. The system 100 further includes a display 110 for displaying an indication of a position of the introducer 104 based on of the impedance measured by the impedance meter 108. In the example shown, the impedance meter 108 and display 110 are part of a single unit 112 (e.g. a single handheld or desktop unit).

Referring now to FIG. 2, in the example shown, the puncture device 102 is a mechanical puncture device, and includes elongate shaft 114 (also referred to herein as a "puncture device shaft") extending between a proximal end 116 (also referred to herein as a "puncture device proximal end") and a distal end 118 (also referred to herein as a "puncture device distal end"). A puncturing tip 120 is at the distal end, for puncturing tissue. In the example shown, the puncturing tip 120 is a sharp tip. The sharp tip can be, for example, beveled, pyramidal, or conical (as shown). A hub 122 (also referred to herein as a "puncture device hub") is at the proximal end 116, for handling and manipulating the puncture device 102.

In alternative examples, the puncture device can be a radiofrequency (RF) puncture device. In such examples, the puncturing tip can include an RF puncture electrode, and the system can include an RF generator to which the RF puncture electrode is electrically connectable for the delivery of RF energy to the RF puncture electrode. In such examples, the distal end of the puncture device can be blunt.

Referring still to FIG. 2, in the example shown, the introducer 104 includes elongate shaft 124 (also referred to herein as an "introducer shaft") extending between a proximal end 126 (shown in FIG. 3, and also referred to herein as an "introducer proximal end") and a distal end 128 (also referred to herein as an "introducer distal end"). The distal end 128 of the shaft 104 is generally blunt, in order to avoid

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damaging tissue. A hub 130 (also referred to herein as an "introducer hub") is at the proximal end 126 of the shaft 124, for handling and manipulating the introducer 104.

Referring to FIGS. 3 and 4, the introducer 104 includes a lumen 132 extending therethrough, from the proximal end 126 of the shaft 124 to the distal end 128 of the shaft 124. The puncture device 102 (not shown in FIGS. 3 and 4) is advanceable through the lumen 132, from the proximal end 126 of the shaft 124 to the distal end 128 of the shaft 124, to position the puncturing tip 120 of the puncture device 102 proud of the distal end 128 of the shaft 124.

Referring still to FIGS. 3 and 4, as mentioned above, the introducer includes an electrode 106, which is at the distal end 128 of the shaft 124 and is distally facing. More specifically, in the example shown, the shaft 124 includes an electrically conductive tube 134 (e.g. a steel tube or other metallic tube), and an electrically insulative sheathing 136 (e.g. a high-density polyethylene sheathing or other polymeric sheathing) on the tube 134. At the distal end 128, the tube 134 is electrically exposed, and the electrically exposed section of the tube 134 forms the electrode 106. The remainder of the tube 134 provides an electrical conductor that extends proximally from the electrode 106, for electrically connecting the electrode 106 to the impedance meter 108, via an insulated wire 138.

In alternative examples, the electrode can be of another configuration. For example, the electrode can be a stand-alone metallic element at or near the distal end of the shaft, and a wire or other electrical conductor can extend proximally from the electrode for connecting the electrode to the impedance meter.

Referring now to FIGS. 5 and 6, in use, the introducer 104 can be advanced percutaneously towards the pericardium 140 of the heart. For example, the introducer 104 can be advanced via the subxiphoid approach, with the puncture device (not shown in FIGS. 5 and 6) shrouded within the lumen 132 (not visible in FIGS. 5 and 6) in order to prevent tissue coring, or extending slightly proud of the introducer to facilitate passage through adipose tissue. As the introducer 104 is advanced, the impedance associated with the electrode 106 (i.e. the impedance of an electrical circuit of which the electrode 106 is a part) can be measured by the impedance meter 108, to assess the position of the introducer 104. More specifically, referring first to FIG. 5, as the introducer 104 is advanced, the impedance meter 108 can continuously measure the impedance associated with the electrode 106. When the electrode 106 (not visible in FIG. 6) contacts the pericardium 140 of the heart, as shown in FIG. 6, the impedance will change, and this change can be used as an indication of the position of the electrode 106 relative to the pericardium, which can in turn be used as an indication of the position of the distal end 128 of the introducer 104. As the introducer 104 is advanced, the impedance meter 108 can communicate with the display 110, and based on the impedance, the display 110 can display an indication of the position of the introducer 104. For example, as shown in FIGS. 5 and 6, the display 110 can display an indication of whether or not the electrode 106 is in contact with the pericardium 140 of the heart. This can be done by displaying text, or by a change in color, or by illumination or a light, or by the emission of a sound. Alternatively, the display 110 can display raw impedance measurements, which can then be interpreted by a user in order to determine whether or not the electrode 106 is in contact with the pericardium 140 of the heart.

When the display 110 indicates that the introducer 104 is in contact with the heart, the method can proceed. That is,

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the puncture device **102** can be advanced out of the lumen **132**, to puncture the pericardium.

In an alternative example (not shown), the electrode can be on the puncture device. For example, in the case of a mechanical puncture device, the electrode can be part of the puncturing tip. In the case of an RF puncture device, the RF electrode can also serve as the electrode for impedance measurement. The system can then measure impedance to assess the position of the puncture device. More specifically, as the puncture device passes through the pericardium, the impedance associated with the electrode will change and can be measured to assess the position of the puncturing tip, to determine whether the puncturing tip has punctured the pericardium. The display can then display an indication of whether the puncture device has punctured the pericardium.

As mentioned above, in the example shown, the impedance meter and display are part of a single unit. In alternative examples, the impedance meter and display may be provided in separate units. For example, the impedance meter can be built into the hub of the introducer, and the display can be provided as a standalone display.

In any of the above examples, the puncture device and/or the introducer can include one or more radiopaque markers, to facilitate viewing under fluoroscopy.

While the above description provides examples of one or more processes or apparatuses or compositions, it will be appreciated that other processes or apparatuses or compositions may be within the scope of the accompanying claims.

To the extent any amendments, characterizations, or other assertions previously made (in this or in any related patent applications or patents, including any parent, sibling, or child) with respect to any art, prior or otherwise, could be construed as a disclaimer of any subject matter supported by the present disclosure of this application, Applicant hereby rescinds and retracts such disclaimer. Applicant also respectfully submits that any prior art previously considered in any related patent applications or patents, including any parent, sibling, or child, may need to be re-visited.

We claim:

1. A system for pericardial puncture comprising:

a medical assembly comprising,

an introducer having an elongate shaft extending between a proximal end and a distal end and defining a single lumen therethrough, the elongate shaft having a longitudinal axis extending between the proximal end and the distal end, the elongate shaft comprising:

an electrically conductive metallic tube;

a polymeric, electrically insulative sheathing on the metallic tube, the insulative sheathing having a tapered portion surrounding a distal portion of the electrically conductive metallic tube;

the introducer having a distally facing electrode at the distal end of the shaft, the electrode comprising a distal end portion of the metallic tube that is electrically exposed through the polymeric insulative sheathing, the electrode encircling the lumen and being coaxial with the longitudinal axis of the elongate shaft, and an elongated puncture device disposed within the lumen and extendable from the distal end of the shaft;

an impedance meter electrically connectable with the electrode for measuring an impedance associated with the electrode; and

a display for displaying an indication of a position of the medical assembly device, wherein the indication of the position is based on of the impedance.

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2. The system of claim **1**, wherein the indication of the position of the medical assembly is an indication of whether the medical is in contact with a target tissue.

3. The system of claim **1**, wherein the shaft comprises an electrical conductor extending proximally from the electrode for electrically connecting the electrode to the impedance meter.

4. The system of claim **1**, wherein

the puncture device is a radiofrequency puncture device; and

the system further comprises a radiofrequency generator electrically connectable with the electrode for delivering radiofrequency energy to the electrode.

5. The system of claim **1**, wherein the puncture device comprises a sharp tip.

6. The system of claim **1**, wherein the indication of a position of the medical assembly device includes an indication of whether or not the electrode is in contact with a pericardium of the heart.

7. The system of claim **6**, wherein the indication of whether or not the electrode is in contact with the pericardium of the heart is displayed text.

8. A method for pericardial puncture, comprising:

a. advancing an introducer towards a pericardium, the introducer having a distal end with a distalmost tip, the introducer having an elongate shaft, the elongate shaft having a longitudinal axis extending between the proximal end and the distal end, the elongate shaft comprising: an electrically conductive metallic tube; a polymeric, electrically insulative sheathing on the metallic tube, the insulative sheathing having a tapered portion surrounding a distal portion of the electrically conductive metallic tube; the introducer including a distally facing electrode at the distalmost tip, the electrode comprising a distal end portion of the metallic tube that is electrically exposed through the polymeric insulative sheathing, and the electrode being coaxial with the longitudinal axis of the elongate shaft;

b. advancing a puncture device through the introducer towards the pericardium, the puncture device including a puncturing tip having a puncturing tip electrode;

c. puncturing the pericardium with the puncture device; and

d. during at least one of step a., step b., and step c., measuring an impedance to assess a position of the introducer and a position of the puncture device, the impedance associated with the distally facing electrode and the puncturing tip electrode.

9. The method of claim **8**, wherein in step d., the position of the distal end of the introducer is assessed.

10. The method of claim **8**, wherein step d. comprises determining whether the distal end of the introducer is in contact with the pericardium.

11. The method of claim **8**, wherein step d. comprises determining whether the puncturing tip has punctured the pericardium.

12. The method of claim **8**, further comprising:

e. displaying an indication of the position of the introducer and/or the position of the puncture device.

13. The method of claim **12**, wherein step e. comprises displaying an indication of whether the introducer has contacted the pericardium, wherein the indication includes text, illumination, or light.

14. The method of claim **12**, wherein step e. comprises displaying an indication of whether the puncture device has punctured the pericardium, wherein the indication includes text, illumination, or light.

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